

50-cycle characterisation across three manufacturers, 8× cycling reduction vs pure

PyBaMM

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Second-life repurposing of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited cycling. Pure PyBaMM [1] reaction-limited SEI calibration requires ≈ 400 cycles per cell for < 3 pp SoH extrapolation on used cells; prior work [3, 4] improves calibration quality but not the cycling budget. *Can one physics-informed neural network with fixed hyperparameters cut required characterisation across LFP manufacturers?*

We train a joint PINN across all cells in a cohort: $SoH_{\theta}(n) = SoH_{init} - softplus(NN_{\theta})$ with hard-monotonic decrement and per-cell latent embedding. Physics constraint: $dSoH/dn = -k_{SEI}(i) SoH^{p(i)}$ (Ramadass-canonical rxn-limited SEI [5]) with per-cell learnable (k_{SEI}, p) warm-started from a linear fit on the training window. Loss combines data MSE, physics MSE on collocation points across the *full* cycle domain, and boundary/monotonicity regularisation. Adam + cosine, 10k epochs, ≈ 4 min on a GPU.

Applied to 20 LFP cells across three manufacturers (7 MFR_A, 9 MFR_C, 4 MFR_B) with *identical* architecture and hyperparameters (K=50, ≈ 68 k params): the joint PINN achieves held-out RMSE < 3 pp SoH on **18/20 cells** (Fig. 1). On deep-second-life MFR_A cells (15–25 pp fade over 1000+ cycles), where pure physics fails catastrophically (median 20 pp), the PINN maintains **median 1.93 pp** — a **10× reduction** and beats physics 7/7 head-to-head (Fig. 2). On near-fresh MFR_B/MFR_C cells (1–3 pp fade), physics is near-optimal (< 0.4 pp) and PINN matches without harm. **Versus the pure-physics K=400 baseline, the same PINN reaches equivalent accuracy at K=50 — 8× cycling reduction with make-agnostic transfer.**

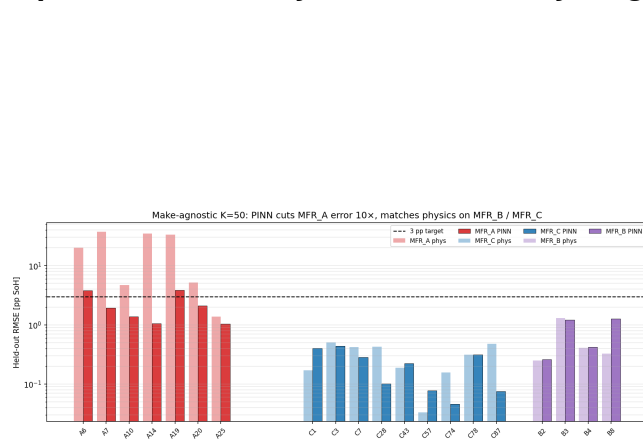


Figure 1: Per-cell held-out RMSE across 3 manufacturers (log scale). MFR_A physics bars tower above the 3 pp target; PINN matched or below across all makes.

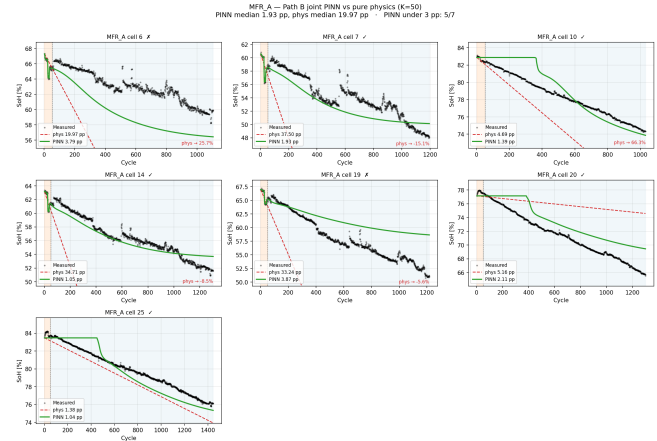


Figure 2: MFR_A trajectories (K=50). Physics dashed lines exit the frame with impossible-SoH annotations; PINN green tracks measured.

Results hold for LFP at 25°C isothermal; cells flagged by an upstream shape filter (non-monotonic, three-regime, batch-transition data artefacts) remain out of scope. PyBaMM v26.4.3's Prada2013 rxn-limited SEI rate is embedded as the physics constraint.

References. [1] V. Sulzer et al., *Python Battery Mathematical Modelling (PyBaMM)*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO₄-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025 Book of Abstracts, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025 Book of Abstracts, p. 12. [5] P. Ramadass et al., *First-principles capacity fade model for Li-ion cells*, J. Electrochem. Soc. 151(2), A196–A203 (2004).